

Appendix

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1 Additional Background

The Food Stamp Program (FSP) was first established under the Johnson administration in 1964 as part of the War on Poverty, though various predecessor programs stretch back to the New Deal.

The central component of SNAP is the provision of monthly food vouchers, with some restrictions on the types of food allowed. Benefits are delivered through Electronic Benefit Cards (EBT), which are essentially debit cards which have a recipient's monthly benefits prepaid onto them. While EBT has existed since the 90's, it was not universally used by all states until 2004. Before that point, some states still used physical stamps.

Benefit amounts, referred to as allotments, depend primarily on income and household size, though the type of income, as well as certain living expenses, can also affect them. As the name might suggest SNAP benefits are designed to be supplemental, paying for a portion of a household's food while the household is expected to contribute 30% of its net income to purchasing its food. As a result, SNAP benefits make up the difference between 30% of a household's monthly net income, and the amount the household would need to spend on food each month under the USDA's Thrifty Food Plan, which is a set of official benchmarks of the necessary level of food expenditure for households with various compositions. However, SNAP also provides a token minimum benefit for households whose income is high enough to reduce monthly benefits to 0 but too low to lose eligibility.

2 Additional Theory

Those at point D, referred to as group 1, are program incumbents who are complying with the ABAWD requirement by working 20 hours per week (in 1B only 17 of those hours are paid) but some of whom would have preferred to remain on the program while working and earning less. In practice this group also contains those working a bit more than 20 because the shifts you need to

hit 20 hours will often go a bit over 20 ie: 3 shifts of 8 hours. The General Work Requirement may sometimes partially constrain their ability to reduce work hours on the extensive margin when the ABAWD requirement is removed. I expect it to unambiguously decrease, or not increase, labor supply on the intensive margin, and maybe on the extensive margin if general req doesn't bind. For the aging out of the ABAWD requirement, incumbents who were working may quit their jobs in response to the income effect produced by being allowed to keep their benefits while working less. However, as discussed below, this is constrained by their still being subject to the General requirement. As discussed in Han 2022, the substitution effect from the benefit phase out is likely to have less of an impact on them as they already experience this aspect of the program, and so the change in their incentives experienced at 50 is modest. For incumbents not already working when they turned 50, they may respond to these incentives by reducing job seeking. If they would have found a job in the next month if subject to the work requirement, and now will not do this, then the reduced job seeking will reduce their potential employment outcome going forward. This effect will be muted if the reasons they faced persistent barriers to employment. Working incumbents who choose to continue to work after 50 may respond similarly in the event that they lose their job. For non-incumbents who apply after turning 50, similar effects are at play. For those working when they turned 50, the income effect may lead them to quit. This could potentially take the form of individuals reducing their labor supply in order to reduce their income enough to qualify for benefits. However, it is questionable whether there are very many individuals who would view this trade-off as beneficial. Additionally, those who were working may quit due to the benefit reduction formula's substitution effect, which is likely to be larger than for working incumbents as it is newly being introduced to them as they join SNAP. However, as Han 2022 points out, the shape of the benefit reduction formula will make this not applicable in many cases. For those who were not already working when they applied, while they have no employment to reduce, the removal of the ABAWD requirement may reduce employment in the long run by reducing job seeking incentives by providing them with SNAP income. However, they may also increase their employment. If a lack of financial resources or food acted as a barrier to employment then they may now be more able to find work (a variant of the poverty trap model). Additionally, joining SNAP at 50 will make someone subject to the General Work requirement which, while not requiring work, does do various things to encourage work. Some of these things might be successful and produced increased employment.

For those who were at F when they turned 50, while they have no employment to reduce, receiving SNAP income after the removal of the ABAWD requirement may reduce employment in the long run by reducing job seeking incentives. If they would have found a job in the next month if subject to the work requirement, and now will not do this, then the reduced job seeking will reduce their potential employment outcome going forward. Working incumbents (group 1) who choose to continue to work after 50 may respond similarly in the

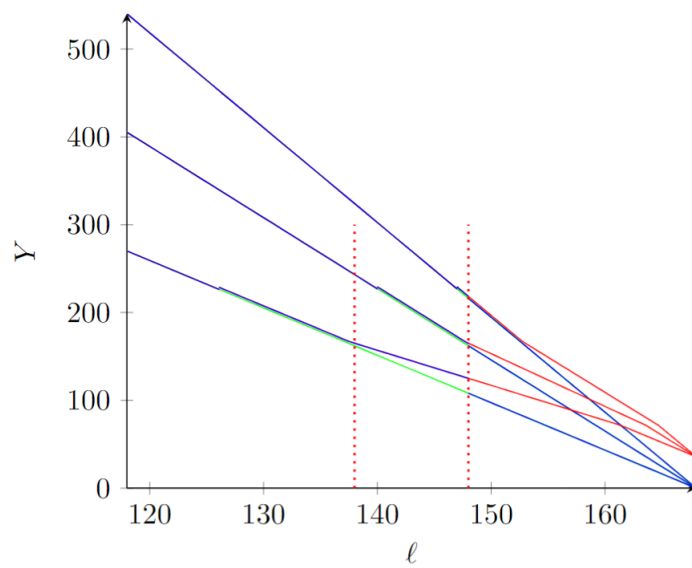
event that they lose their job. However, joining SNAP may also increase long run employment for those at F as it will make them subject to the General Work requirement which encourages work and provides employment resources which, if successful, may produce increased employment. For those working when they turned 50, the income effect may lead them to quit. While this would likely be concentrated in those on EMF, this could potentially take the form of individuals on AB reducing their labor supply in order to reduce their income enough to qualify for benefits. However, as is discussed in depth below, it is questionable whether there are very many individuals who would view this trade-off as beneficial. Additionally, those on EMF who join SNAP may quit due to the benefit reduction formula's substitution effect, which is likely to be larger than for incumbents as it is newly imposed on them. However, as Han 2022 points out, the shape of the benefit reduction formula will make this not applicable for those moving from MF to GH.

In a similar vein, potential discontinuous changes in SNAP E&T can be seen as either a source of bias or as part of the treatment of aging out of the work requirements. SNAP E&T is intended for those under 60, and it generally designed to prioritize finding work opportunities for those under 50 (insert citation), and so the availability of E&T services are likely to change suddenly at the ages of 50 or 60. We can think of E&T as component of the General Requirement, and thus think of this removal as a component of the treatment of aging out of the requirement. Similarly, we can choose to think of E&T's prioritization of those subject to the ABAWD requirement as an extension of this requirement. If E&T provides employment, or increases the probability that an individual finds employment (which is not certain), then its cessation will decrease the expected value employment going forward for incumbents. For non-incumbents who did not work before joining after 60, lowered job seeking/opportunities would lower their potential employment. If E&T only acts as an ordeal mechanism to deter program entry then any reductions at 50 would increase program entry.

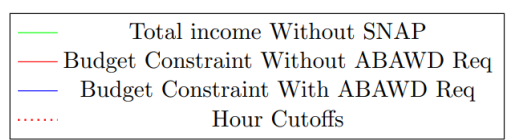
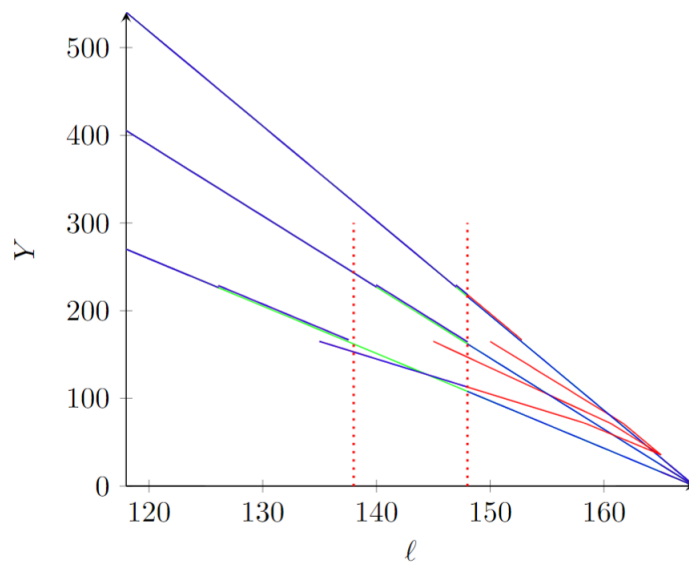
3 Additional Theory Figures

3.1 Figure A1 - The ABAWD Requirement

A1A

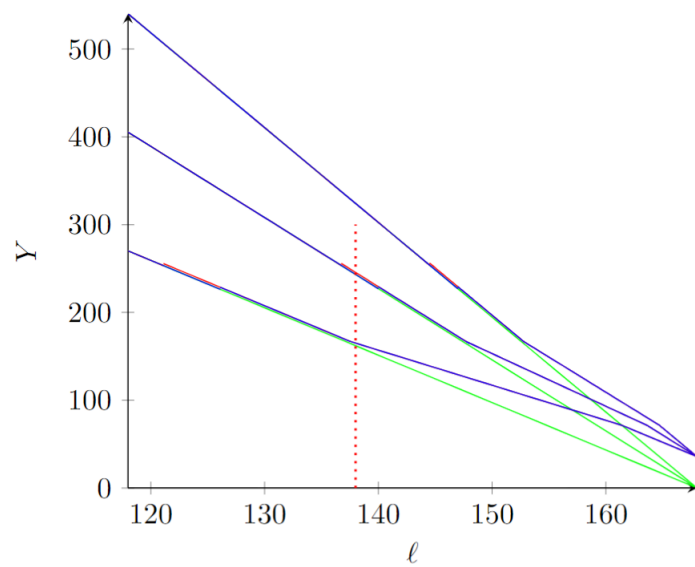


A1B

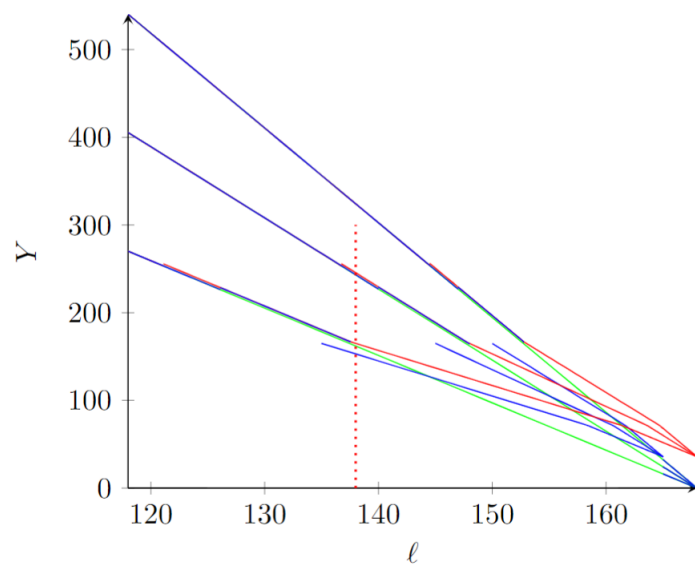


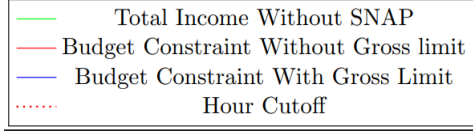
3.2 Figure A2A - The General Requirement

A2A



A2B





Figures A1 and A2 are analogous to figures 1 and 2 from the main paper. They are intended to extend figures 1 and 2 to a larger number of wage rates. From highest to lowest blue line, each graph depicts the budget constraints for someone earning twice the minimum wage, 1.5 times the minimum wage, and the federal minimum wage.

4 Reasons why I do not find Han 2022's Results to be credible

- Exemption and waiving is based on the geographic level, rather than on whether an individual was actually exempt. It is a continuous variable. This means that the exempt variable is not deterministic of whether someone is exempt. It is also imprecise when it applies relative to observations of the outcome variable. I guess this can be thought of as measurement error in the exemption/waiver variable. This will bias estimates towards 0 if this measurement is random. Hard to know if it could be nonrandom. They are measuring age in years instead of month and also he excludes people who are 50, thus making it a doughnut. This makes their measurement of how far you are from the cutoff be less precise. Thus both parts of their DDD are measured imprecisely. On the flip side, it means we capture the effects a longer period of time away from turning 50 (capturing maybe some of the dynamic effects and reducing downwards bias).
- Probably doesn't drop everyone who was really disabled. This would cause dilution. Drops people living with an elderly person for some reason (I should double check why). In general, drops based on disability, having kids or elderly in your hh. Only includes people under 300% fpl.
- His DDD estimate is the difference between being under 50 in an exempt area and being over 50 in an exempt area. However, given that an area is exempt, we wouldn't expect this to have any effect. Maybe it could be interpreted as the additional difference that being exempt makes when you are under 50? However it never has a separate variable for estimating the effect of being under 50 so would it still have this effect? Would the fixed effects compensate for this? At the very least, it isn't specifically the effect of adding or removing the work req. in an area. Ok I've thought about it and I think that it would genuinely give a DDD effect representing the additional effect that exemption has if you are under 50. I'm not sure

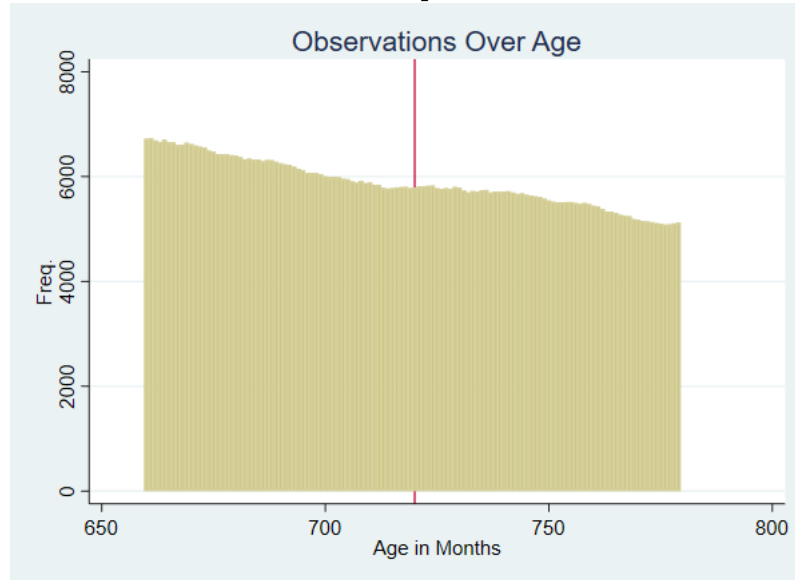
if this would capture the effect of being under 50 in general, and I'm not sure about exactly what the FE would capture.

- If work requirements cause people to work then including local unemployment would be a bad control for employment, and maybe also for participation, because it would be a mechanism through which exemption affects outcomes. Also, the local unemployment rate is redundant to the year by location fe.
- The participation variable is based on whether someone was participating at any point in the past year, even if it was for just a short time before getting kicked off. Similar things apply to the employment outcome. These can be thought of as measurement errors, overstating the level of employment and participation. Given that it is imprecise when they occur in time relative to exemption, this measurement error may be compounded by measurement error in exemption.
- Goodman-Bacon issues
- There would be sample selection based on income correlated with exemptions and unemployment. If a recession causes people to loose their jobs then this will mean that those people are more likely to show up in the sample, thus driving down employment.
- Substantial under-reporting of SNAP participation in ACS would hurt their precision
- It doesn't captured the reduction in the treatment effect from those who were not disabled before 50 and then became disabled after 50.

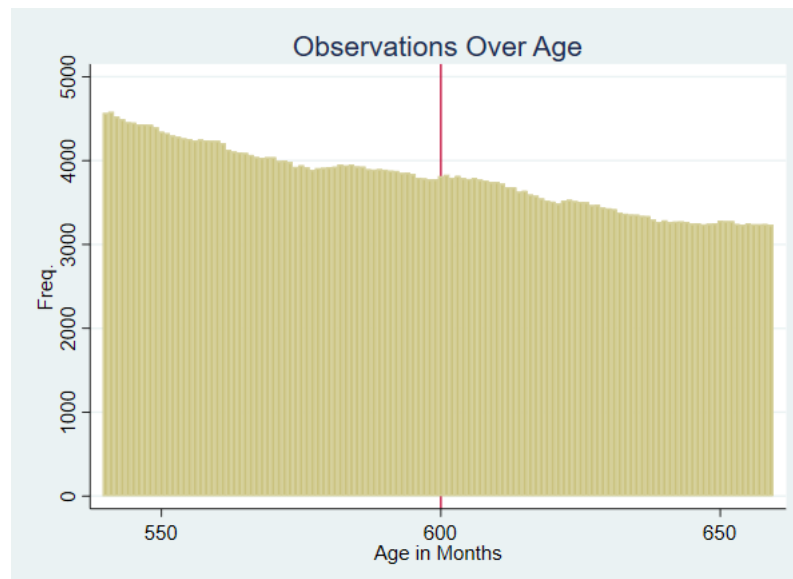
5 Additional Descriptive Figures

5.1 Figure A3 - Observations Over Age

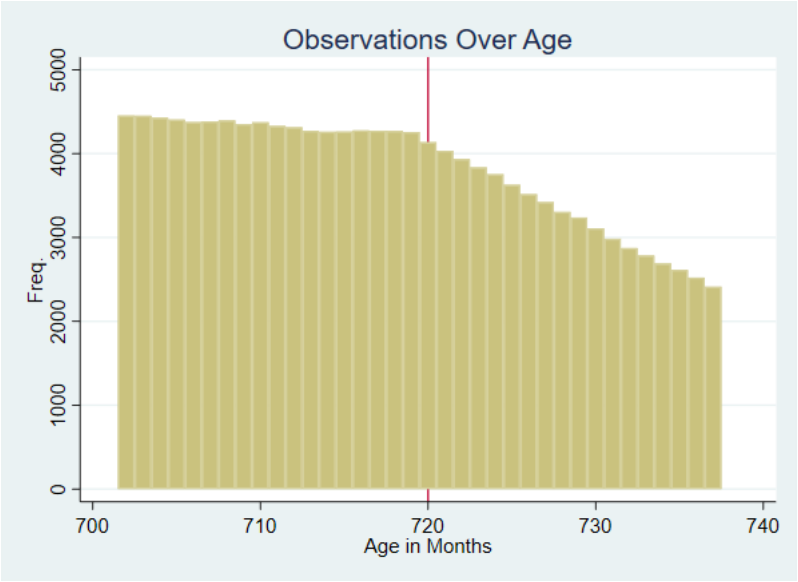
A3A - Combined General Sample



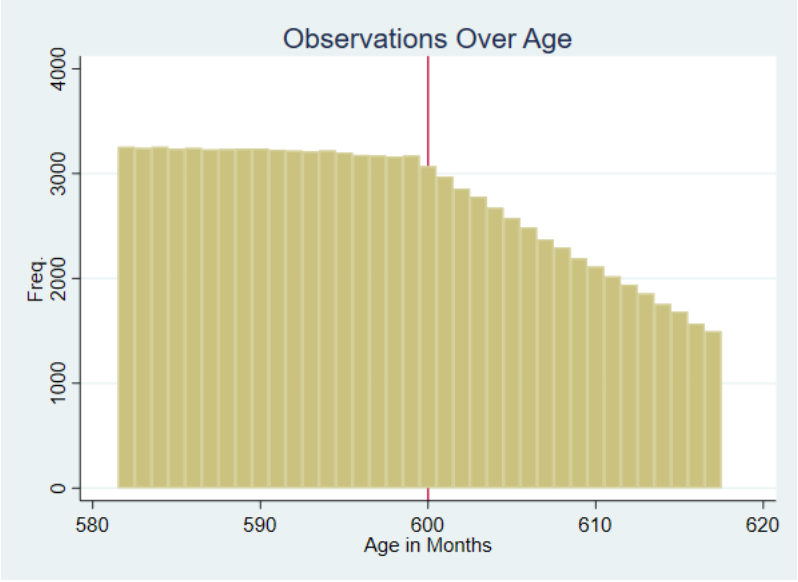
A3B - Combined ABAWD Sample



A3C - Stock General Sample



A3D - Stock ABAWD Sample



6 Robustness Checks

To evaluate the robustness of my estimates, the following robustness checks rerun equations 1 through 5 using an alternate set of samples defined based on education. The primary sample is selected based on individuals who spent > 10% of months 250% of the federal poverty line who ever lived in a family where at least one person was receiving SNAP. Here I constructed analogous stock (unrestrictive or UR) and combined (unfiltered or UF) work requirements for each work requirement to include observations from individuals whose highest educational attainment was a high school degree or less, at some point in the SIPP.

6.1 RD

Table A1A

FE Regression Discontinuities (SNAP Participation Equation 1)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.004*** (0.001)	-0.002 (0.001)	-0.002 (0.001)	-0.001 (0.001)
Disabled	0.024*** (0.002)	0.045*** (0.003)	0.021*** (0.003)	0.043*** (0.007)
Family Exemption	0.020*** (0.006)	0.001 (0.004)	0.059*** (0.010)	0.039*** (0.010)
Children		0.013*** (0.002)		0.009*** (0.003)
UI Benefits	0.013*** (0.003)	-0.002 (0.002)	0.017*** (0.003)	0.003 (0.003)
Constant	0.078*** (0.004)	0.049*** (0.003)	0.025*** (0.006)	0.016*** (0.002)
Observations	274,761	181,684	77,405	59,508
R-squared	0.008	0.005	0.010	0.005
Number of u_lgtkey	12,942	9,858	4,750	3,970
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A1B

Non-FE Regression Discontinuities (SNAP Participation Equation 2)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	-0.001 (0.005)	0.001 (0.005)	-0.002 (0.004)	-0.001 (0.005)
Disabled	0.199*** (0.007)	0.260*** (0.011)	0.101*** (0.016)	0.108*** (0.026)
Family Exemption	0.094*** (0.023)	0.107*** (0.024)	0.059 (0.037)	0.048 (0.041)
Children		0.018*** (0.005)		0.022*** (0.005)
UI Benefits	0.027*** (0.009)	0.011 (0.008)	0.043*** (0.015)	0.015 (0.011)
Constant	0.082*** (0.008)	0.065*** (0.008)	0.049*** (0.009)	0.043*** (0.009)
Observations	274,761	181,684	77,405	59,508
R-squared	0.134	0.167	0.041	0.030
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1				

Table A1C

Cattane et al. Regression Discontinuities (SNAP Participation Equation 3)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
RD_Estimate	-0.002 (0.006)	-0.003 (0.006)	-0.000 (0.004)	0.001 (0.005)
Observations	274,761	181,684	77,405	59,508
Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1				

Table A2A

FE Regression Discontinuities (Equation Employment 1)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.003 (0.003)	0.002 (0.002)	-0.003* (0.002)	0.010*** (0.002)
Disabled	-0.289*** (0.007)	-0.432*** (0.006)	-0.346*** (0.008)	-0.478*** (0.010)
Family Exemption	-0.229*** (0.008)	-0.338*** (0.010)	-0.323*** (0.018)	-0.422*** (0.028)
Children		0.013*** (0.003)		0.004 (0.006)
Constant	0.636*** (0.006)	0.787*** (0.008)	0.771*** (0.014)	0.894*** (0.011)
Observations	274,761	181,682	77,405	59,506
R-squared	0.105	0.118	0.103	0.107
Number of u.lgtkey	12,942	9,858	4,750	3,970
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A2B

Non-FE Regression Discontinuities (Equation Employment 2)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.012* (0.007)	-0.002 (0.007)	0.006 (0.009)	0.016* (0.009)
Disabled	-0.702*** (0.005)	-0.834*** (0.005)	-0.662*** (0.012)	-0.692*** (0.024)
Family Exemption	-0.599*** (0.021)	-0.654*** (0.028)	-0.713*** (0.012)	-0.752*** (0.032)
Children		0.002 (0.006)		-0.006 (0.009)
Constant	0.696*** (0.012)	0.807*** (0.010)	0.704*** (0.020)	0.804*** (0.017)
Observations	274,761	181,682	77,405	59,506
R-squared	0.422	0.530	0.138	0.156

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A2C

Cattane et al. Regression Discontinuities (Employment Equation 3)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
RD_Estimate	-0.010 (0.009)	0.005 (0.009)	-0.003 (0.012)	0.011 (0.010)
Observations	274,761	181,682	77,405	59,506
Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1				

6.2 2WFE

Table A3A

2-Way FE (SNAP Participation Equation 4)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.004 (0.004)	-0.002 (0.004)	-0.002 (0.003)	-0.001 (0.004)
Disabled	0.024*** (0.005)	0.045*** (0.009)	0.021** (0.010)	0.043*** (0.016)
Family Exemption	0.020 (0.014)	0.002 (0.015)	0.059** (0.028)	0.039 (0.035)
Children		0.013** (0.005)		0.009 (0.006)
UI Benefits	0.013* (0.007)	-0.002 (0.005)	0.017* (0.010)	0.003 (0.007)
Constant	0.077*** (0.005)	0.030*** (0.007)	0.024*** (0.006)	0.005 (0.009)
Observations	274,761	181,684	77,405	59,508
R-squared	0.008	0.005	0.010	0.005
Number of u_lgtkey	12,942	9,858	4,750	3,970
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1				

Table A3B

2-Way FE (Employment Equation 4)				
VARIABLES	(1) UF. Gen.	(2) UF. ABAWD	(3) UR. Gen.	(4) UR. ABAWD
Over Age Threshold	0.004 (0.005)	0.002 (0.006)	-0.004 (0.008)	0.010 (0.008)
Disabled	-0.288*** (0.010)	-0.432*** (0.017)	-0.347*** (0.021)	-0.478*** (0.033)
Family Exemption	-0.228*** (0.018)	-0.338*** (0.026)	-0.324*** (0.054)	-0.422*** (0.054)
Children		0.013* (0.007)		0.004 (0.012)
Constant	0.627*** (0.009)	0.817*** (0.011)	0.762*** (0.018)	0.945*** (0.024)
Observations	274,761	181,682	77,405	59,506
R-squared	0.103	0.118	0.103	0.107
Number of u_lgtkey	12,942	9,858	4,750	3,970
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

6.3 DID

Table A4A

Between-Variation DID (SNAP Participation Equation 5)					
VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	-0.015 (0.013)	-0.002 (0.013)	-0.012 (0.009)	-0.008 (0.010)	0.003 (0.006)
exemption_strata_inter	0.014 (0.014)		0.012 (0.010)		
exemption_strata	-0.146*** (0.014)		-0.067*** (0.016)		
Disabled	0.092*** (0.010)	0.110*** (0.015)	0.230*** (0.014)	0.226*** (0.018)	0.097*** (0.017)
Family Exemption	0.015 (0.030)	0.046 (0.042)	0.107*** (0.027)	0.107*** (0.036)	0.123** (0.049)
UI Benefits	0.030*** (0.009)	0.043*** (0.014)	0.015 (0.009)	0.014 (0.012)	0.046*** (0.013)
exemption_strata2_inter		0.003 (0.014)		0.007 (0.010)	
exemption_strata2		-0.127*** (0.018)		-0.065*** (0.018)	
Children			-0.031* (0.016)	-0.033* (0.018)	0.022*** (0.005)
Pre 2008 x Threshold					0.003 (0.007)
Pre 2008					-0.040*** (0.006)
Constant	0.215*** (0.015)	0.203*** (0.019)	0.130*** (0.017)	0.132*** (0.020)	0.091*** (0.009)
Observations	250,643	99,324	165,014	66,231	100,573
R-squared	0.161	0.166	0.179	0.174	0.037
Time FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A4B

Within-Variation DID (SNAP Participation Equation 5)					
VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	0.032*** (0.010)	0.015 (0.009)	0.005 (0.009)	0.005 (0.008)	-0.000 (0.007)
exemption_strata_inter	-0.037*** (0.011)		-0.013 (0.009)		
Disabled	0.024*** (0.007)	0.022** (0.009)	0.030*** (0.011)	0.043*** (0.015)	0.039*** (0.013)
Family Exemption	0.033** (0.017)	0.072*** (0.025)	0.002 (0.018)	0.024 (0.032)	0.014 (0.027)
UI Benefits	0.012* (0.006)	0.023** (0.010)	-0.001 (0.005)	0.005 (0.008)	0.003 (0.009)
exemption_strata2_inter		-0.018* (0.010)		-0.007 (0.009)	
Children			0.008 (0.008)	0.025** (0.010)	0.011** (0.005)
Pre 2008 x Threshold					-0.000 (0.008)
Constant	0.076*** (0.005)	0.077*** (0.009)	0.036*** (0.007)	0.028** (0.012)	0.023*** (0.009)
Observations	250,643	99,324	165,014	66,231	100,573
R-squared	0.008	0.008	0.003	0.005	0.004
Number of u_lgtkey	11,858	6,086	9,020	4,423	6,541
Time FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A5A

Between-Variation DID (Employment Equation 5)					
VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	-0.010*** (0.003)	-0.021*** (0.005)	-0.024*** (0.007)	-0.004 (0.009)	-0.006 (0.009)
exemption_strata_inter	-0.095*** (0.008)		0.002 (0.010)		
exemption_strata	0.354*** (0.014)		0.110*** (0.011)		
Disabled	-0.446*** (0.013)	-0.496*** (0.016)	-0.773*** (0.009)	-0.783*** (0.012)	-0.693*** (0.016)
Family Exemption	-0.412*** (0.022)	-0.424*** (0.038)	-0.615*** (0.030)	-0.628*** (0.045)	-0.672*** (0.045)
exemption_strata2_inter		-0.034*** (0.009)		-0.004 (0.012)	
exemption_strata2		0.294*** (0.017)		0.097*** (0.015)	
Children			0.082*** (0.008)	0.072*** (0.012)	-0.003 (0.008)
Pre 2008 x Threshold					-0.003 (0.011)
Pre 2008					0.040*** (0.010)
Constant	0.408*** (0.016)	0.461*** (0.021)	0.732*** (0.012)	0.734*** (0.018)	0.760*** (0.013)
Observations	250,643	99,324	165,014	66,229	100,571
R-squared	0.443	0.457	0.547	0.534	0.166
Time FE	YES	YES	YES	YES	YES
Robust standard errors in parentheses					
*** p<0.01, ** p<0.05, * p<0.1					

Table A5B

Within-Variation DID (Employment Equation 5)					
VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	0.038*** (0.006)	0.026*** (0.007)	-0.001 (0.008)	0.011 (0.009)	-0.002 (0.011)
exemption_strata_inter	-0.041*** (0.008)		0.013 (0.011)		
Disabled	-0.277*** (0.013)	-0.300*** (0.018)	-0.406*** (0.020)	-0.425*** (0.029)	-0.443*** (0.023)
Family Exemption	-0.230*** (0.020)	-0.257*** (0.032)	-0.339*** (0.029)	-0.372*** (0.040)	-0.388*** (0.041)
exemption_strata2_inter		-0.044*** (0.007)		-0.002 (0.010)	
Children			0.025** (0.010)	-0.005 (0.012)	0.013 (0.010)
Pre 2008 x Threshold					0.012 (0.013)
Constant	0.647*** (0.010)	0.659*** (0.015)	0.810*** (0.012)	0.871*** (0.023)	0.902*** (0.017)
Observations	250,643	99,324	165,014	66,229	100,571
R-squared	0.067	0.093	0.087	0.104	0.098
Number of u_lgtkey	11,858	6,086	9,020	4,423	6,541
Time FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1